



American Society for Gastrointestinal Endoscopy guideline on the role of endoscopy in acute lower gastrointestinal bleeding

Nauzer Forbes, MD, MSc, FASGE,^{1,2} Nayantara Coelho-Prabhu, MD, FASGE,³ Hana Osman, MD,⁴ Howard Guo, MD,^{1,2} Sunil Samnani, MBBS,⁵ Indraneel Datta, MD,⁶ Edward Wolfgang Lee, MD, PhD,⁷ Anirudh Mirakhur, MD,⁸ Wasif M. Abidi, MD, PhD,⁹ Natalie Cosgrove, MD,¹⁰ Madhav Desai, MD, MPH,¹¹ Sara Ghoneim, MD,¹² Calvin Lee, MD, MPH, FASGE,¹³ Jorge D. Machicado, MD, MPH, FASGE,¹⁴ Jared Magee, DO, MPH,^{15,16} Neil B. Marya, MD,¹⁷ Saowanee Ngamruengphong, MD, FASGE,¹⁸ Michael D. Rice, MD,¹⁴ Wenly Ruan, MD,¹⁹ Monica Saumoy, MD, MS,²⁰ Sunil G. Sheth, MD, FASGE,²¹ Nikhil R. Thiruvengadam, MD,²² Bashar J. Qumseya, MD, MPH, FASGE,²³ Nirav C. Thosani, MD, MHA, ASGE Standards of Practice Committee Chair¹¹

Calgary, Alberta, Canada; Rochester, Minnesota; Los Angeles, California; Houston, Texas; Orlando, Florida; Boston, Massachusetts; New Hyde Park, New York; Ann Arbor, Michigan; Bethesda, Maryland; Worcester, Massachusetts; Baltimore, Maryland; Plainsboro, New Jersey; Loma Linda, California; Gainesville, Florida, USA

Disclaimer: This guideline document was prepared by the Standards of Practice Committee of the American Society for Gastrointestinal Endoscopy using the best available scientific evidence and considering a multitude of variables, including, but not limited to, adverse events, patients' values, and cost implications. The purpose of these guidelines is to provide the best practice recommendations that may help standardize patient care, improve patient outcomes, and reduce variability in practice. We recognize that clinical decision-making is complex. Guidelines, therefore, are not a substitute for a clinician's judgment. Such judgments may, at times, seem contradictory to our guidance because of many factors that are impossible to fully consider by guideline developers. Any clinical decisions should be based on the clinician's experience, local expertise, resource availability, and patient values and preferences. This document is not a rule and should not be construed as establishing a legal standard of care, or as encouraging, advocating for, mandating, or discouraging any particular treatment. Our guidelines should not be used in support of medical complaints, legal proceedings, and/or litigation because they were not designed for this purpose.

This clinical practice guideline from the American Society for Gastrointestinal Endoscopy provides evidence-based recommendations regarding the role of endoscopy in the diagnosis and management of acute lower GI bleeding (LGIB). This document was developed using the Grading of Recommendations Assessment, Development, and Evaluation (GRADE) framework. In the broad context of acute LGIB, this guideline addresses the use of colonoscopy versus CT +/- angiography as the first modality, urgent versus nonurgent colonoscopy, and prepped versus unprepped colonoscopy. For diverticular bleeding in particular, this guideline assesses endoscopic band ligation versus clipping and direct clipping of the bleeding site or vessel versus indirect clip closure of the diverticulum. All of these clinical questions were answered using the GRADE framework. The guideline also discusses the roles of upper endoscopy and surgery in the setting of acute LGIB and outlines the absolute and relative contraindications of colonoscopy. Finally, the guideline provides an overview of available endoscopic modalities to treat LGIB, with special attention to the evidence supporting the use of specific modalities according to lesion type. A summary of recommendations is provided in Table 1. (Gastrointest Endosc 2026;■:1-12.)

(footnotes appear on last page of article)

INTRODUCTION

Acute lower GI bleeding (LGIB) is generally defined as bleeding that occurs from anywhere in the cecum, colon,

or rectum and typically presents as either hematochezia or, less commonly, melena. Annual hospitalization rates for major bleeding events range between 20 and 40 per 100,000 persons in the United States and Europe.¹⁻⁴ Contemporary data

demonstrate considerably higher rates of major LGIB in patients with relevant comorbidities and those on antithrombotic medications.⁵⁻⁸

Among the various etiologies of LGIB, diverticular disease is the most common, accounting for about 30% of presentations for LGIB.^{9,10} Anorectal pathology, colonic neoplasms, and various colitides are the next most common causes, each accounting for roughly 20% of presentations for major bleeding.⁹ Less common etiologies include iatrogenic LGIB after endoscopic or surgical interventions and vascular lesions such as angioectasia.

Unlike upper GI bleeding (UGIB), in which timely and sometimes emergent management is frequently necessary, the natural history of LGIB is highly variable, ranging from presentations with spontaneous cessation of bleeding to those with ongoing bleeding and hemodynamic instability requiring urgent or emergent intervention. The mortality associated with LGIB has been reported to range between 2% and 4%, with advanced age, coagulopathy, and concomitant comorbidities being associated with a higher risk of death, among other covariates.^{9,11} Several risk scores have been developed and validated for the purposes of distinguishing between those at higher risk of adverse outcomes from LGIB,¹² in part leading to the proposal of several triaging strategies by multiple guideline societies in the assessment of such patients.¹³⁻¹⁵

Colonoscopy is often performed for both the diagnosis and management of acute LGIB. However, a thorough understanding of the indications and contraindications for endoscopic interventions is crucial, particularly in the context of alternative available modalities such as CT angiography or expectant management, any of which could variably be appropriate for a particular patient within a given clinical setting.

As such, the aim of this guideline is to provide high-quality, clinically relevant, evidence-based recommendations regarding the role of endoscopy in the diagnosis and management of acute LGIB, in addition to details regarding endoscopic timing, preparation, specific techniques and modalities, and alternatives, including radiologic and surgical interventions. Other considerations, such as risk stratification, medical management, anticoagulation management, and transfusion strategy, are not systematically reported.

SUMMARY OF METHODS

This document was prepared by the Standards of Practice (SOP) Committee of the American Society for Gastrointestinal Endoscopy (ASGE) and was carried out in accordance with the ASGE Clinical Practice Guideline development policy¹⁶ while following the Grading of Recommendations Assessment, Development, and Evaluation

(GRADE) framework and recommendations.¹⁷ A summary of the methodology used for this document is provided in the checklist of the aforementioned ASGE policy,¹⁶ whereas detailed methodology specific to this guideline is outlined in the [Supplementary Materials](#), available online at www.giejournal.org.

Clinically important outcomes within the context of LGIB were proposed *a priori* by the lead guideline authors and SOP chair and were refined and finalized by all members of the SOP Committee. These included mortality, recurrent bleeding, adverse events (AEs), need for transfusion, need for radiologic intervention, need for surgery (all within 30 days of presentation), length of hospitalization, diagnostic yield, and ability to offer initial hemostasis. With these outcomes in mind, this guideline aimed to answer the following comparative questions using the GRADE process: (1) In patients with acute LGIB, should colonoscopy or CT +/− angiography be used as the initial diagnostic and/or therapeutic modality? (2) In patients with acute LGIB who require colonoscopy, should urgent or nonurgent colonoscopy be performed to diagnose and/or treat the bleeding source? (3) In patients with acute LGIB who require colonoscopy, should prepped or unprepped colonoscopy be performed to diagnose and/or treat the bleeding source? (4) In patients with active or recent diverticular bleeding identified endoscopically, should band ligation or clipping be performed to treat the bleeding source? (5) In patients with active or recent diverticular bleeding identified endoscopically, should direct clipping or indirect clipping be performed to treat the bleeding source?

In addition to the above GRADE-based questions, we also aimed to answer the following questions after systematically reviewing available evidence but not using the full GRADE process: (1) In patients with acute LGIB, when should upper endoscopy be performed in addition to colonoscopy and/or other investigations? (2) In patients with acute LGIB, when should surgery be considered? (3) In patients with acute LGIB, what are the possible modalities of endoscopic treatment according to bleeding lesion type?

Full details of search design and execution, data extraction, data analysis, evidence synthesis, and GRADE panel process are provided in the [Supplementary Materials](#), available online at www.giejournal.org. A summary of recommendations can be found in [Table 1](#).

EXTERNAL REVIEW

The guideline was reviewed by the *Gastrointestinal Endoscopy* Editorial Board, the Sustainable Endoscopy Task Force, and the ASGE Governing Board and made available for public comment on the ASGE website for 30 days, from July 15 to August 15, 2025.

INITIAL ASSESSMENT

Evaluation and risk stratification

As with any acute presentation, patients presenting with suspected LGIB should be evaluated and risk stratified to guide appropriate resuscitative efforts, diagnostic workup, and management. Thorough evaluation should include a full medical history and physical examination, with special attention paid to potential risk factors for LGIB and worse outcomes after LGIB, including advanced age, relevant comorbidities, and baseline medications (including antithrombotic agents),⁵ initial hemodynamic stability, and ongoing bleeding.¹⁸

Several risk scores have been developed to aid in the stratification of patients presenting with suspected LGIB and in the prediction of adverse outcomes including mortality.¹⁹ The test performance characteristics of these have been summarized elsewhere.²⁰ Among these, the Oakland¹² and SHA₂PE²¹ scores were specifically developed to aid in the identification of patients for whom discharge from acute care was safe and for whom outpatient investigations were safe and appropriate. These scores are undoubtedly useful in risk-stratifying patients with suspected LGIB but should never replace clinical judgment.

Resuscitation

In general, patients with hemodynamic instability should be observed and treated in a monitored setting and resuscitated with intravenous crystalloid fluids to normalize heart rate and blood pressure, ideally before any further diagnostic and/or therapeutic modalities. Patients should have 2 large-bore intravenous catheters (or equivalent) placed to appropriately support adequate resuscitation efforts. Transfusion of packed red blood cells is appropriate for those with hematochezia and a drop in their baseline hemoglobin with hemodynamic instability. Typically, a restrictive transfusion strategy is recommended for most patients with LGIB without a history of established cardiovascular disease, with a hemoglobin threshold of 7 g/L.^{14,15} In those with a history of cardiovascular disease or with LGIB amid acute coronary syndrome, a more liberal strategy may be appropriate, although the evidence is not clear enough to define an optimal management strategy for this population.²²

Exclusion of a proximal bleeding source

In patients presenting with hematochezia, what is the role of upper endoscopy?

Recommendation: Upper endoscopy should be performed for all patients presenting with hematochezia and initial hemodynamic instability to rule out an upper GI source unless there is direct imaging evidence of LGIB.

The results of 637 initial articles were screened and informed the review for this question. There are currently no studies directly comparing outcomes with and without upper endoscopy in the setting of acute LGIB. In a randomized controlled trial (RCT) of 85 patients presenting with suspected LGIB (hematochezia without hematemesis) and hemodynamic instability (increased heart rate, systolic blood pressure <90 mm Hg, a decrease in systolic blood pressure >40 mm Hg from baseline, or orthostatic blood pressure changes) or rapid hemoglobin drop (transfusion requirements of at least 4-6 units of packed red blood cells within the first 24 hours), an upper GI source was identified in 15% of patients who underwent upper endoscopy performed within 6 hours of presentation, with peptic ulcers and esophagogastric varices being the most common responsible etiologies identified.²³⁻²⁵ Other retrospective studies have also reported the performance of upper endoscopy in patients presenting with suspected acute LGIB without findings on initial colonoscopy. From these, the yield of subsequent upper endoscopy is similar, ranging between 11% and 22%.²⁶⁻²⁸ In addition to hemodynamic instability, other risk factors for UGIB in patients presenting with hematochezia include male sex, a known history of peptic ulcer disease or cirrhosis, and an elevated blood urea nitrogen-to-creatinine ratio of over 30.²⁸ An alternative approach to upper endoscopy for stable patients with hematochezia is urgent or emergent capsule endoscopy, which has been shown to have reasonable ability to identify bleeding lesions in this setting.^{26,29}

DIAGNOSTIC TESTS

Colonoscopy versus CT for initial diagnosis

In patients presenting with acute LGIB, should colonoscopy or CT +/- angiography be used as the initial diagnostic and/or therapeutic modality?

Recommendation: In patients with acute LGIB, either CT with contrast +/- angiography or colonoscopy is acceptable as an initial diagnostic modality. The ASGE suggests colonoscopy for patients who are stable on presentation or after initial resuscitation. The ASGE suggests contrast CT +/- angiography in patients with ongoing hemodynamic instability (conditional recommendation, very low quality of evidence).

Eight retrospective cohort studies were included in our meta-analysis comparing CT +/- angiography (the "intervention") with colonoscopy (the "comparator") as the initial modality for patients presenting with acute LGIB. Forest plots and evidence summaries for this question are provided in the [Supplementary Materials](#), available

online at www.giejournal.org. In general, there were significant differences in the underlying patient populations receiving CT versus colonoscopy, given the retrospective observational designs of the source studies, suggestive of potential selection bias. Patients with acute LGIB receiving CT +/- angiography as the initial diagnostic modality were older and had more comorbidities (apart from lower rates of acute kidney injury and/or chronic kidney disease), presented with lower blood pressures, and received more blood transfusions.

On meta-analysis of 8 retrospective cohort studies, we found that patients undergoing initial CT +/- angiography had higher mortality, with an odds ratio (OR) for 30-day mortality of 12.64 (95% CI, 2.97-53.75; $I^2 = 0\%$), in addition to higher requirement for blood transfusion, with an OR of 2.29 (95% CI, 1.56-3.35; $I^2 = 0\%$), and longer hospital stays (standardized mean difference [SMD], 2.09 days; 95% CI, 0.62-3.56; $I^2 = 43\%$), compared with patients who had colonoscopy first. There were no differences between CT +/- angiogram and colonoscopy in their diagnostic yield (OR, 1.44; 95% CI, 0.74-2.82; $I^2 = 87\%$) or in their ability to provide initial hemostasis via either direct endoscopic or interventional radiologic approaches (OR, 0.77; 95% CI, 0.25-2.42; $I^2 = 80\%$).

The panel emphasized the importance of being cautious in interpreting the available data comparing CT +/- angiography and the colonoscopy, given several limitations. The most obvious was the retrospective cohort design used for all available studies, making any comparative data in this context prone to bias, including that of preselection of the most severe presentations and/or most comorbid patients for a nonendoscopic approach. Other limitations raised included a proportion of input data being from non-U.S. settings (low generalizability), some of the source studies include patients from the distant past (outdated data), and inconsistency in adjusting for covariates. The panel noted similar diagnostic yields and ability to provide initial hemostasis (with rates of 19%-29%) but acknowledged that the AE rates associated with each technique are relatively unexplored in a comparative manner, with severe confounding present in which comparisons are available.

Given these considerations, the panel ultimately agreed that colonoscopy was an appropriate initial diagnostic and/or therapeutic modality for stable patients, including those who are initially unstable but respond to standard initial resuscitative measures, whereas CT angiogram (followed by radiologically guided intervention, if positive) was likely appropriate for those who were either hemodynamically unstable despite resuscitation or unable to tolerate a full bowel preparation (either orally or via nasogastric tube). The panel opted against using strict criteria to label a patient as stable versus unstable because they felt there is no single criterion or validated score or combination

thereof that can replace clinical judgment. They also felt that neither blood cell transfusion (requirement thereof or number of units) nor ongoing clinical signs of bleeding during admission should guide the decision to pursue one modality over another. Finally, the panel also acknowledged that the choice of one intervention over another can also be influenced by local availability and expertise. This recommendation is incorporated into the overall clinical care algorithm presented in [Figure 1](#).

With regard to the presence of either acute or chronic kidney disease, the radiologists on the panel were clear that in the case of severe acute LGIB, this should not be a deterrent to performing CT +/- angiography in many patients with modest kidney disease. The most challenging cases, however, arise in patients with very poor renal function who could conceivably require dialysis with any further deterioration. The panel also noted that expectant management (ie, no endoscopic or other intervention) may also be appropriate for patients presenting with stable acute LGIB without transfusion requirement, especially in instances in which there has been a recent high-quality colonoscopy completed with an explanation for the patient's presentation. For example, it may be reasonable to observe a patient presenting with stable hematochezia in the context of having undergone a complete colonoscopy with adequate bowel preparation 6 months prior, with the only finding being sigmoid diverticulosis.

Nuclear medicine modalities

The availability and performance of CT modalities have reduced the need for nuclear medicine evaluation in LGIB.³⁰ The role of scintigraphic investigation in LGIB is now largely limited to settings in which CT or endoscopy are either not readily available or inconclusive.

MANAGEMENT

Colonoscopy versus CT +/- embolization for management

The comparison of colonoscopy and CT is discussed in the previous section, "Colonoscopy versus CT for Initial Diagnosis," and includes considerations related to management. In general, CT +/- embolization is the preferred modality for patients with ongoing hemodynamic instability or for borderline unstable patients who have positive findings on a diagnostic CT. It is important to note that the likelihood of successful hemostasis increases with the shortest possible window between a positive CT and angiography. Furthermore, patients in under-resourced settings may lack access to either urgent colonoscopy or urgent interventional radiology, which can influence initial management decisions.

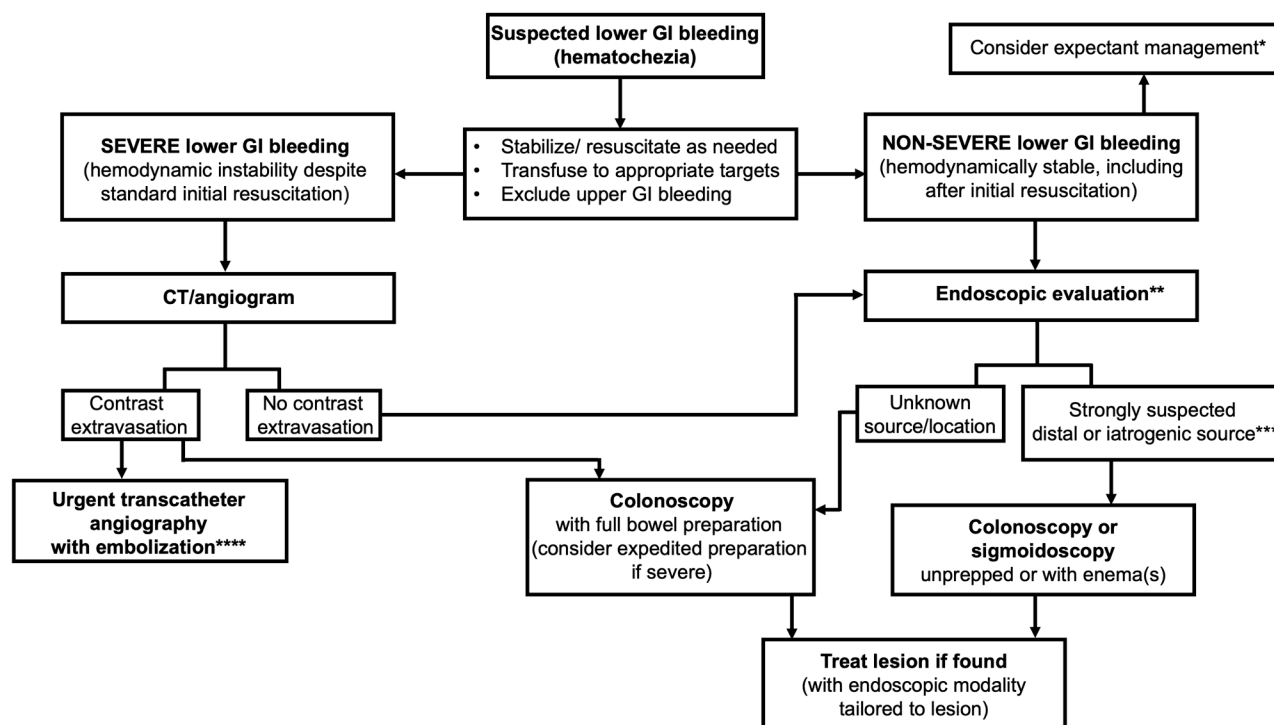


Figure 1. Clinical decision-making algorithm for patients presenting with acute lower GI bleeding. *Examples might include patients with stable bleeding and a recent high-quality colonoscopy showing only diverticulosis. **The timing of endoscopic evaluation for nonsevere lower GI bleeding has not been associated with improved outcomes. ***Examples might include postpolypectomy/endoscopic submucosal dissection/EMR bleeding, malignancy with known location, known left-sided diverticular disease, and rectal trauma. ****If unavailable or unsuccessful and ongoing severe bleeding, consider urgent colonoscopy with expedited purgative bowel preparation or surgery.

Timing of colonoscopy

In patients presenting with acute LGIB, should urgent or nonurgent colonoscopy be performed to diagnose and/or treat the bleeding source?

Recommendation: In patients with acute LGIB, the ASGE suggests that colonoscopy be performed when the patient is stable enough for the procedure and able to tolerate a full bowel preparation, regardless of timing (conditional recommendation, moderate quality of evidence).

A total of 18 studies were included in our meta-analysis comparing urgent colonoscopy with nonurgent colonoscopy for patients presenting with acute LGIB. Of the 18 studies, 4 were RCTs. Fourteen of the 18 studies (2 of the 4 RCTs) defined "urgent" colonoscopy as one performed within 24 hours of presentation, with widely variable definitions in the remaining studies. Emphasis was heavily placed on results from the RCTs. Forest plots and evidence summaries for this question are provided in the [Supplementary Materials](#), available online at www.giejournal.org. We stratified data by study design when possible and included sensitivity analyses in which we excluded data from studies not using 24 hours as the cut-off between urgent and nonurgent colonoscopy.

On meta-analysis, there was no difference in 30-day mortality with urgent versus nonurgent colonoscopy (OR, 0.76; 95% CI, 0.16-3.50; $I^2 = 0\%$) based on 3 RCTs including 385 patients. There was no difference in recurrent bleeding rates with urgent versus nonurgent colonoscopy (OR, 1.23; 95% CI, 0.46-3.27; $I^2 = 69\%$) based on 4 RCTs including 463 patients. However, in the sensitivity analysis excluding the 2 RCTs not following the 24-hour cutoff, the odds of recurrent bleeding were higher for patients undergoing urgent colonoscopy within 24 hours (OR, 3.05; 95% CI, 1.23-7.56; $I^2 = 0\%$). Patients receiving urgent colonoscopy within 24 hours had a shorter hospital stay (SMD, -0.96 days; 95% CI, -1.46 to -0.47 ; $I^2 = 0\%$). There were no differences between urgent and nonurgent groups in terms of need for surgery (OR, 0.83; 95% CI, 0.20-3.54; $I^2 = 19\%$), need for radiologic intervention (OR, 4.91; 95% CI, 0.81-29.80; $I^2 = 0\%$), transfusion requirements (OR, 1.27; 95% CI, 0.66-2.44 from 1 RCT), or AEs (OR, 1.39; 95% CI, 0.45-4.35; $I^2 = 0\%$). Care was also taken to consider data from the 2 RCTs with different time cutoffs; one administered purgative bowel preparation and performed an urgent colonoscopy within 8 hours, with no differences in outcomes between groups.³¹ Conversely, in the other study, colonoscopy performed within 12 hours of presentation was compared with 36 hours or later and did not result in any meaningful differences in outcomes.²³

The panel felt that, overall, there were subtle differences in outcomes between urgent and nonurgent colonoscopy. The most obvious was recurrent bleeding, which was similar between groups when all RCT data were included but higher in the urgent group when only studies using the 24-hour cutoff were retained. Given these data, the panel concluded that if colonoscopy were to be pursued within the first 24 hours after presentation with acute LGIB, a full bowel preparation should be administered and that the procedure should be performed in a controlled setting with several endoscopic devices and techniques available so that a bleeding lesion could be definitively treated if it were discovered. The approximate reduction in length of hospitalization by 1 day with urgent colonoscopy (along with associated costs) was also considered an important consideration, although the panel was mixed in their opinions on whether urgent colonoscopy (within 24 hours) was broadly acceptable, feasible, or implementable across sites. Specifically, the panel acknowledged that although earlier colonoscopy may reduce length of stay, its implementation can potentially be limited by access to endoscopy and issues with after-hours staffing, particularly in nonacademic settings.

The panel acknowledged that there are conceivable situations in which more urgent (or even unprepped) lower endoscopy could reasonably be pursued, including a known or pre-established bleeding lesion located in the left side of the colon or a strongly suspected iatrogenic source of bleeding (ie, postpolypectomy, post-EMR, post-endoscopic submucosal dissection bleeding, or recent rectal tube) at a known and easily identifiable location within the colon (see algorithm in Fig. 1).

Colonoscopy preparation

In patients presenting with acute LGIB, should prepared or unprepared colonoscopy be performed to diagnose and/or treat the bleeding source?

Recommendation: In patients with acute LGIB, the ASGE recommends colonoscopy with full bowel preparation unless there is strong suspicion of either a distal bleeding source or iatrogenic bleeding with known location (strong recommendation, very low quality of evidence).

There was very limited available evidence informing this question. A retrospective cohort study of 194 patients who received either full polyethylene glycol-based bowel preparation or enemas after presenting with hematochezia showed that patients receiving full bowel preparations benefited from both higher diagnostic yield and ability to provide initial hemostasis and, as a result, a decreased need for repeat colonoscopy.³²

Despite the relative lack of evidence addressing this question, the panel felt strongly that a prepared colonoscopy should be the standard for patients presenting with acute LGIB. The panel agreed that the only exceptions to this (see the algorithm in Fig. 1) should occur when a patient has either

(1) a strongly suspected distal source of bleeding (based on prior endoscopy and/or imaging during the current presentation) or (2) an iatrogenic source of bleeding (ie, postpolypectomy, post-EMR, postendoscopic submucosal dissection bleeding, or suspected rectal tube trauma/ulceration) at a known and easily identifiable location within the colon. The panel also noted that one exception to this framework is established radiation proctopathy, which, while a suspected distal source, is generally treated with argon plasma coagulation (APC) and, therefore, should still occur only after a full bowel preparation given the risk of colonic explosion with APC use with higher levels of intestinal gases.³³

Interventions for diverticular bleeding (I)

In patients with active or recent diverticular bleeding identified endoscopically, should band ligation or clipping be performed to treat the bleeding source?

Recommendation: In patients with active or recent diverticular bleeding identified endoscopically, the ASGE suggests band ligation or clipping based on the location of bleeding, clinical expertise, and device availability. For transverse colonic or more proximal lesions, the ASGE suggests band ligation when feasible. For distal lesions, the ASGE suggests clipping (conditional recommendation, very low quality of evidence).

A total of 7 comparative studies were included in our meta-analysis, informing recommendations for this question (all observational; 6 retrospective and 1 prospective). Of note, much of the available literature assessing this question studied the use of band ligators on gastroscopes. Colonoscope-specific band ligation devices have been studied in a limited manner³⁴ but have not yet become widely available at the time of this guideline's writing. All studies reported recurrent bleeding rates, and comparative data from 2329 patients showed that the odds of this outcome were lower with banding than with clipping (OR, 0.33; 95% CI, 0.23-0.47; $I^2 = 24\%$). However, in the studies with available subgroup data, banding was associated with decreased recurrent bleeding only for right-sided lesions, with no apparent benefits for left-sided lesions.^{35,36} There was also a lower need for radiologic intervention with banding compared with clipping (OR, 0.43; 95% CI, 0.24-0.76; $I^2 = 0\%$) from 3 studies including 1887 patients. Although not significant, there was also a trend toward decreased transfusion requirements with banding compared with clipping (OR, 0.73; 95% CI, 0.52-1.02; $I^2 = 20\%$) based on 4 studies including 2022 patients. There was no difference in the need for surgery for refractory or ongoing bleeding (OR, 1.08; 95% CI, 0.36-3.23; $I^2 = 0\%$).

Although not significant, there was a trend toward increased AEs with banding compared with clipping (OR, 2.41; 95% CI, 0.70-8.24; $I^2 = 0\%$). AEs observed in patients receiving band ligation included perforation (which required subsequent surgery 100% of the time), diverticulitis, and

fever, whereas AEs observed in patients receiving clipping included diverticulitis. It was noted that all available studies addressing this comparison were carried out in Asia, limiting generalizability to North American settings. Furthermore, most subjects included in the analysis were recruited as part of the largest study by Kobayashi et al³⁵ a retrospective analysis of 1679 patients. Forest plots and evidence summaries for this question are provided in the [Supplementary Materials](#), available online at www.giejournal.org.

Although the panel acknowledged the lower rates of both recurrent bleeding and radiologic intervention with banding compared with clipping, there were some reservations expressed regarding the potential for AEs. Overall, the panel concluded that the balance of favorable versus unfavorable effects generally favored the practice of banding. However, given that the risk reduction associated with banding was driven by right-sided lesions in the studies with available subgroup data, with no apparent benefits for left-sided lesions,^{35,36} the panel concluded that recommendations for banding at this time should be limited to right-sided colonic lesions (defined as transverse colon or more proximal lesions) and only when feasible. The panel also expressed reservations regarding the widespread feasibility of banding in North American patients with acute LGIB, with the main concerns relating to inexperience and possible technical challenges of navigating a scope loaded with a banding device into the right side of the colon, especially in patients with higher body mass index. Furthermore, banding requires removal of the colonoscope entirely to load the banding device, followed by reinsertion into the bleeding site with the bander loaded. This can be technically challenging in an angulated or redundant colon in general, but especially amid active bleeding. Reidentification of the bleeding site can be aided by the initial placement of a marking clip and/or tattoo, with the advantage of a tattoo being that it will permanently remain at the bleeding site, which could help guide future endoscopic and/or surgical management. Therefore, the panel concluded that banding of diverticular lesions should be attempted only when a provider is experienced in this approach and only when the appropriate banding device is available (ie, one that can be loaded onto a colonoscope rather than a gastroscope, which is not readily available in North America at present).

Interventions for diverticular bleeding (II)

In patients with active or recent diverticular bleeding identified endoscopically in whom clipping is chosen, should direct clipping or indirect clipping be performed to treat the bleeding source?

Recommendation: In patients with active or recent diverticular bleeding identified endoscopically in whom clipping is chosen, the ASGE suggests direct clipping over indirect clipping whenever feasible, particularly in the proximal colon (conditional recommendation, very low quality of evidence).

When endoscopically treating an actively bleeding diverticulum or one with a nonbleeding visible vessel or other stigmata of recent bleeding with clips, there are 2 main techniques an endoscopist can use. The first, the indirect clipping method, involves closing the entire diverticular orifice. Depending on the size of the diverticulum, this can sometimes require “zipper” closure with multiple clips, similar to endoscopic closure of postpolypectomy defects. Conversely, direct clipping involves the direct visualization of a vessel (either within the dome of a diverticulum or along its lip) and the subsequent application of 1 or more clips directly to the vessel to treat the bleeding source. Small-mouthed diverticula can make deploying clips within a diverticulum difficult or impossible.

We identified 9 comparative studies (all observational, 1 prospective) containing data addressing direct versus indirect clipping. There was no difference between direct and indirect clipping in their ability to provide initial hemostasis (OR, 1.17; 95% CI, 0.43-3.20; $I^2 = 20\%$). In terms of mortality, there was no difference with direct versus indirect clipping (OR, 3.39; 95% CI, 0.27-42.21; $I^2 = 32\%$) among the very limited number of patients ($n = 132$) across studies with this outcome assessed. Direct clipping was associated with significantly lower odds of recurrent bleeding compared with indirect clipping (OR, 0.58; 95% CI, 0.44-0.77; $I^2 = 0\%$). Direct clipping was also associated with a lower requirement for radiologic intervention (OR, 0.50; 95% CI, 0.25-0.97; $I^2 = 0\%$) but not with a lower requirement for surgery (OR, 1.03; 95% CI, 0.12-8.68; $I^2 = 0\%$). There were no differences with respect to AEs (OR, 1.27; 95% CI, 0.25-6.46; $I^2 = 0\%$) or transfusion requirements (OR, 0.88; 95% CI, 0.42-1.82; $I^2 = 42\%$). As with the previous question, all studies were conducted in Asia, with much higher proportions of right-sided diverticular lesions than in Western settings. Forest plots and evidence summaries for this question are provided in the [Supplementary Materials](#), available online at www.giejournal.org.

Unlike the question of banding versus clipping of diverticular lesions, the panel was more comfortable suggesting direct clipping over indirect clipping, despite similar benefits primarily in right-sided lesions driving most of the results. The primary reason was that there was no difference in AEs between the 2 techniques. The panel clearly noted situations in which direct clipping might not be feasible, including small-mouthed diverticula and lack of visibility of a clear vessel or bleeding source to target. Other considerations include choosing the right size and type of clip tailored to the lesion in question. As with all endoscopic interventions, the panel noted that experience and familiarity with direct clipping should ideally be gained under initial supervision before attempting the technique independently.

Summary of endoscopic management options by bleeding type

In patients presenting with acute LGIB, what are the options for endoscopic management?

Recommendation #1. In patients with acute diverticular LGIB, band ligation or clipping is suggested depending on the location of the lesion, clinical expertise, and device availability. For right-sided colonic lesions, when feasible, band ligation is suggested, although through-the-scope clipping is also a viable option if band ligation is not feasible. For left-sided lesions, through-the-scope clipping is suggested. When clipping is used, direct clipping is suggested over indirect clipping, when feasible.

Recommendation #2. In patients with postresection acute LGIB, we suggest through-the-scope clipping or thermal coagulation as initial modalities, with hemostatic powder considered for adjunctive use or for refractory bleeding despite initial endoscopic therapy.

Recommendation #3. In patients with acute LGIB from vascular lesions such as angiodysplasias or radiation proctopathy, we suggest APC.

Recommendation #4. In patients with acute malignant LGIB, we suggest the use of hemostatic powder or the Nd:YAG laser as measures to achieve temporary hemostasis, either with palliative intent or before definitive surgical management.

Diverticular bleeding. For patients with bleeding diverticular lesions, recommendations are summarized in the previous section, "Interventions for diverticular bleeding." In addition to the discussed modalities, snare ligation has also been studied in a limited fashion in this context.³⁷ Over-the-scope clipping has not been studied extensively in the context of acute diverticular bleeding, but early evidence appears promising in support of this modality.³⁸ Thermal modalities can be considered for diverticular bleeding, but caution must be exercised given the thin-walled anatomy of diverticula and the risk of perforation, especially at the base of the diverticulum. Therefore, risks associated with the use of thermal modalities in diverticular bleeding can be minimized if applied at the rim of diverticula (as opposed to the base) and by the use of lower power and duration of coagulation.³⁹ Evidence supporting the use of topical hemostatic agents, including novel therapies such as self-assembling peptide gels in the treatment of acute diverticular bleeding, is limited.⁴⁰ Importantly, although Doppler probes have been assessed for their ability to endoscopically risk-stratify lesions causing UGIB,⁴¹ data are lacking to guide the use of these devices in the setting of LGIB. However, given that the type of underlying vessel can be predictive of recurrent bleeding and overall natural history in LGIB,⁴² further research on the use of Doppler probes in this setting

is likely warranted, ideally via prospective comparative trials including patients presenting with various etiologies of LGIB.

Postresection bleeding. Delayed bleeding can occur after routine polypectomy, EMR, or endoscopic submucosal⁴³ dissection and typically presents within 14 days but can rarely present up to the 30-day mark. Predictors of active bleeding at the time of colonoscopy include left-sided lesions, pedunculated lesions, and anticoagulant use.⁴⁴ In a decision modeling analysis, repeat colonoscopy for suspected delayed postpolypectomy bleeding was beneficial in identifying and treating a bleeding lesion in 22% of cases, with a number needed to treat of 4.5 patients.⁴⁵ Again, head-to-head trials are lacking within this context. Well-established endoscopic therapies for such patients include through-the-scope clip application, which has been proven effective,⁴⁶ and contact and noncontact thermal therapies, although high-quality evidence supporting their use is relatively scarce. Thermal therapy must be applied cautiously to a resection bed because of the risk of perforation, particularly in the right side of the colon. Adjunctive epinephrine injection can be used as a temporizing measure, whereas hemostatic powders have been studied in a limited fashion for this indication, with successful hemostasis achieved when used as a monotherapy, as part of a combination therapy, or as a rescue therapy.⁴⁷

Bleeding vascular lesions. Although vascular lesions most often present with occult blood loss and resultant anemia, acute overt LGIB is also possible, with advanced patient age, cardiac comorbidities, anticoagulant use, multiple lesions, and small lesions all associated with a higher risk of active bleeding.⁴⁸ No direct head-to-head trials have been carried out to compare modalities in the treatment of these lesions. Noncontact thermal therapy in the form of APC is the most common modality used to treat bleeding vascular lesions. In a large series of patients undergoing APC for both actively bleeding and recently bleeding angiodysplasias, the rate of recurrent bleeding after therapy was 7.4%.⁴⁹ It is important for endoscopists to have a working knowledge of electrocautery unit settings in this context because pulsed APC modes offer more superficial level of cauterization than continuous modes. To further mitigate the risk of perforation in the thinner walled right side of the colon, submucosal injection before APC therapy can be considered. A single small case series demonstrated successful eradication of vascular lesions with a technique of submucosal injection and then resection (achieved with the same technique as EMR) followed by submucosal coagulation of the exposed feeding vessel.⁵⁰ Radiation-associated vascular ectasia (radiation proctopathy), which can cause significant hematochezia, can also be considered a vascular lesion, and successful cessation of bleeding after APC has also been described for this population.^{51,52}

Malignant bleeding. Limited data exist assessing available modalities in the treatment of acute LGIB from

malignancy. Endoscopic therapy in this context should be viewed as temporizing only and, therefore, performed as either a bridge to definitive surgical management or as palliative therapy for symptom management. Use of the Nd:YAG laser in this context has been described⁵³ but with limited evidence to support its effectiveness and seemingly waning current availability in Western settings. Hemostatic powder has also been assessed in this context via case series and demonstrates promise in achieving immediate temporary hemostasis,⁵⁴ which, given its wider availability, appears to have made this modality the mainstay of endoscopic treatment for malignant bleeding.

The role of surgery

In patients presenting with acute LGIB, what is the role of surgery?

Recommendation: Surgery should be considered a last resort for patients with acute LGIB, being performed only when bleeding is either not amenable to endoscopic or radiologic intervention or when both these interventions have failed to control bleeding.

With the advancement of radiologic and endoscopic technologies and techniques, the role of surgery in acute LGIB is much more limited than previously reported. As an example, in a cohort study of 1112 patients with acute LGIB, a decrease in surgery from 22.6% to 16.6% was demonstrated between the periods 1988 to 1997 and 1998 to 2006,⁵⁵ with even smaller numbers of patients requiring surgery in more recent, smaller observational cohorts of patients with acute LGIB.^{56,57} In a nationwide prospective United Kingdom cohort of 2528 patients consecutively seen over 2 months in 2015 for acute LGIB, only 0.2% ultimately required surgery.¹⁰

When 26 patients with acute LGIB treated with surgery after positive CT angiography findings were compared with 62 patients treated by interventional radiology in the same cohort, the surgical patients were younger and had higher base excesses and lactate levels.⁵⁸ Although the length of hospitalization was similar between groups, AE rates were higher with surgery than with radiologic intervention.⁵⁸ In a separate retrospective cohort of 1614 patients who underwent emergency surgery for acute refractory LGIB, 30-day mortality was high at 12.2%. Older age, functional dependence, and higher American Society of Anesthesiologists score increased the odds of short-term mortality, whereas a laparoscopic approach and higher hematocrit at the time of surgery were associated with lower mortality.⁵⁹

Therefore, although rare, surgery may still be required for LGIB when all other interventions have failed, an identified bleeding source is not amenable to intervention by radiologic or endoscopic means, or, in exceptional cases, when patients are too unstable for any other intervention.

Additional rare cases that would potentially prompt emergent laparotomy as the initial modality of choice for acute LGIB might include an aortocolonic fistula or bleeding Meckel's diverticulum.

SUSTAINABILITY PERSPECTIVE

This guideline was reviewed by the ASGE's Sustainable Endoscopy Task Force. As with all GI conditions, the best interests of the individual patient should be prioritized over sustainability concerns. However, when there is a choice of suitable management options, the ASGE recommends that clinicians consider environmental impact in their decision-making. In general, a practice that follows evidence-based guidelines represents a sustainable practice because it not only provides high-quality and essential care but also reduces unnecessary tests and interventions. Specific to this guideline, the ASGE suggests the following sustainability best practices: (1) avoiding low-yield procedures and duplicative testing wherever possible (eg, foregoing CT scan in patients with stable LGIB and involving the gastroenterology service early in the decision-making process); (2) reducing hospital admissions, unnecessary procedures, and length of stay for low-risk patients using available risk prediction tools, as described in this document²⁰; and (3) aiming to perform definitive endoscopic therapy at the index procedure to avoid repeat procedures, prolonged hospitalizations, and/or readmissions.

CONCLUSIONS AND FUTURE DIRECTIONS

Our guideline summarizes pertinent available evidence on the role of endoscopy in the management of patients presenting with LGIB. In addition to providing guidance on this topic, our guideline also uncovered several gaps representing priority areas for future research in this field, including (1) execution of more RCTs to assess differences in approaches and techniques compared here with minimize confounding; (2) comparative studies adequately powered to assess AEs associated with more novel endoscopic techniques; (3) studies assessing learning curves associated with more novel techniques; (4) cost-effectiveness studies to better understand the impacts associated with various approaches and/or techniques; and (5) studies on currently understudied investigations and interventions in the context of LGIB, including over-the-scope clipping, which has been shown in the context of UGIB to be an effective modality (both as rescue therapy and, increasingly, as primary therapy^{60,61}), band ligation therapy, given the considerations discussed here, Doppler probe-directed therapy, given the considerations discussed here, and the utility of and indications for capsule endoscopy in this setting.

TABLE 1. Summary of recommendations

Recommendation (or best practice advice)	Strength of recommendation	Quality of the evidence
In patients presenting with hematochezia, upper endoscopy should be performed for all patients presenting with hematochezia and initial hemodynamic instability to rule out an upper GI source unless there is direct imaging evidence of LGIB.	N/A	N/A
In patients presenting with acute LGIB, either CT with contrast +/- angiography or colonoscopy is acceptable as an initial diagnostic modality. The ASGE suggests colonoscopy for patients who are stable on presentation or after initial resuscitation. The ASGE suggests contrast CT +/- angiography in patients with ongoing hemodynamic instability.	Conditional	Very low
In patients with acute LGIB, the ASGE suggests that colonoscopy be performed when the patient is stable enough for the procedure and able to tolerate a full bowel preparation, regardless of timing.	Conditional	Moderate
In patients presenting with acute LGIB, the ASGE recommends colonoscopy with full bowel preparation unless there is strong suspicion of either a distal bleeding source or iatrogenic bleeding with known location.	Strong	Very low
In patients with active or recent diverticular bleeding identified endoscopically, the ASGE suggests band ligation or clipping based on the location of bleeding, clinical expertise, and device availability. For transverse colonic or more proximal lesions, the ASGE suggests band ligation when feasible. For distal lesions, the ASGE suggests clipping.	Conditional	Very low
In patients with active or recent diverticular bleeding identified endoscopically in whom clipping is chosen, the ASGE suggests direct clipping over indirect clipping whenever feasible, particularly in the proximal colon.	Conditional	Very low
In patients with acute diverticular LGIB, band ligation or clipping is suggested depending on the location of the lesion, clinical expertise, and device availability. For right-sided colonic lesions, when feasible, band ligation is suggested, although through-the-scope clipping is also a viable option if band ligation is not feasible. For left-sided lesions, through-the-scope clipping is suggested. When clipping is used, direct clipping is suggested over indirect clipping, when feasible.	N/A	N/A
In patients with postresection acute LGIB, we suggest through-the-scope clipping or thermal coagulation as initial modalities, with consideration of hemostatic powder use for adjunctive use or for refractory bleeding despite initial endoscopic therapy.	N/A	N/A
In patients with acute LGIB from vascular lesions such as angiodysplasia or radiation proctopathy, we suggest argon plasma coagulation.	N/A	N/A
In patients with acute malignant LGIB, we suggest the use of hemostatic powder or the Nd:YAG laser as measures to achieve temporary hemostasis, either with palliative intent or before definitive surgical management.	N/A	N/A
In patients presenting with acute LGIB, surgery should be considered a last resort for patients with acute LGIB, being performed only when bleeding is either not amenable to endoscopic or radiologic intervention or when both these interventions have failed to control bleeding.	N/A	N/A

APC, Argon plasma coagulation; ASGE, American Society for Gastrointestinal Endoscopy; LGIB, lower GI bleeding; NA, not applicable.

GUIDELINE UPDATE

ASGE guidelines are reviewed for updates approximately every 5 years or when new data may influence a recommendation. Updates follow the same ASGE guideline development process.

DISCLOSURE

The following authors disclosed financial relationships: N. Forbes: Consultant for Boston Scientific Corporation. N. Coelho-Prabhu: Consultant for Boston Scientific Corporation and Alexion Pharma; research support from Cook Endoscopy and Fujifilm. E. Lee: Consultant for and grants from Gore Medical; consultant for Terumo Medical. A. Mirakhur: Medical advisory board for Boston Scientific

and Abbott Vascular; consultant for Shockwave Medical and Penumbra Medical. W. Abidi: Consultant for Ambu Medical and Boston Scientific Corporation; research support from Morphe Medical. N. Cosgrove: Consultant for Olympus and Boston Scientific Corporation. S. Ghoneim: research funding from Eli Lilly. C. Lee: Stockholder in AbbVie, Inc, Intuitive Surgical, and Moderna, Inc. N. Marya: Consultant for Boston Scientific Corporation. S. Ngamruengphong: Consultant for Boston Scientific Corporation, Olympus, Neptune Medical, and Fujifilm. M. Rice: Consultant for Medtronic and ANX Robotica. M. Saumoy: Consultant for Enterasense. B. Qumseya: Consultant for Medtronic, Inc and Assentio Management, LLC; speaker for Castle Biosciences. N. Thosani: Consultant for PENTAX of America, Inc, Boston Scientific Corporation, and Ambu Medical; royalties from UpToDate; ownership in ROSEAI. All other authors disclosed no financial relationships.

ACKNOWLEDGMENTS

The authors thank Dr Dennis Jensen, Dr Babu Mohan, and Dr John Saltzman for their reviews as representatives of the *GIE* Editorial Board; Dr Jonathan Cohen as the Governing Board Sponsor; and the ASGE Sustainable Endoscopy Task Force for their review. This guideline was funded exclusively by the American Society for Gastrointestinal Endoscopy; no outside funding was received to support the development of this guideline.

REFERENCES

- Lanas A, Perez-Aisa MA, Feu F, et al. A nationwide study of mortality associated with hospital admission due to severe gastrointestinal events and those associated with nonsteroidal antiinflammatory drug use. *Am J Gastroenterol* 2005;100:1685-93.
- Peery AF, Crockett SD, Murphy CC, et al. Burden and cost of gastrointestinal, liver, and pancreatic diseases in the United States: update 2021. *Gastroenterology* 2022;162:621-44.
- Longstreth GF. Epidemiology and outcome of patients hospitalized with acute lower gastrointestinal hemorrhage: a population-based study. *Am J Gastroenterol* 1997;92:419-24.
- Lanas A, García-Rodríguez LA, Polo-Tomás M, et al. Time trends and impact of upper and lower gastrointestinal bleeding and perforation in clinical practice. *Am J Gastroenterol* 2009;104:1633-41.
- Forbes N, Yi Q, Moayyedi P, et al. Incidence and predictors of major gastrointestinal bleeding in patients on aspirin, low-dose rivaroxaban, or the combination: secondary analysis of the COMPASS randomised controlled trial. *Aliment Pharmacol Ther* 2024;60:737-48.
- Giugliano RP, Ruff CT, Braunwald E, et al. Edoxaban versus warfarin in patients with atrial fibrillation. *N Engl J Med* 2013;369:2093-104.
- Cea Soriano L, Lanas A, Soriano-Gabarró M, et al. Incidence of upper and lower gastrointestinal bleeding in new users of low-dose aspirin. *Clin Gastroenterol Hepatol* 2019;17:887-95.e6.
- Miller CS, Dorreen A, Martel M, et al. Risk of gastrointestinal bleeding in patients taking non-vitamin K antagonist oral anticoagulants: a systematic review and meta-analysis. *Clin Gastroenterol Hepatol* 2017;15:1674-83.e3.
- Strate LL, Ayanian JZ, Kotler G, et al. Risk factors for mortality in lower intestinal bleeding. *Clin Gastroenterol Hepatol* 2008;6:1004-10; quiz 955-.
- Oakland K, Guy R, Uberoi R, et al. Acute lower GI bleeding in the UK: patient characteristics, interventions and outcomes in the first nationwide audit. *Gut* 2018;67:654-62.
- Velayos FS, Williamson A, Sousa KH, et al. Early predictors of severe lower gastrointestinal bleeding and adverse outcomes: a prospective study. *Clin Gastroenterol Hepatol* 2004;2:485-90.
- Oakland K, Jairath V, Uberoi R, et al. Derivation and validation of a novel risk score for safe discharge after acute lower gastrointestinal bleeding: a modelling study. *Lancet Gastroenterol Hepatol* 2017;2:635-43.
- Oakland K, Chadwick G, East JE, et al. Diagnosis and management of acute lower gastrointestinal bleeding: guidelines from the British society of Gastroenterology. *Gut* 2019;68:776-89.
- Triantafyllou K, Gkolfakis P, Gralnek IM, et al. Diagnosis and management of acute lower gastrointestinal bleeding: European Society of Gastrointestinal Endoscopy (ESGE) Guideline. *Endoscopy* 2021;53:850-68.
- Sengupta N, Feuerstein JD, Jairath V, et al. Management of patients with acute lower gastrointestinal Bleeding: an updated ACG Guideline. *Am J Gastroenterol* 2023;118:208-31.
- ASGE Standards of Practice Committee; Thosani NC, Desai M, Abidi WM, et al. American Society for Gastrointestinal Endoscopy clinical practice guideline development policy and checklist. *Gastrointest Endosc* 2025;101:261-6.
- Guyatt GH, Oxman AD, Vist GE, et al. GRADE: an emerging consensus on rating quality of evidence and strength of recommendations. *BMJ* 2008;336:924-6.
- Patel P, Nigam N, Sengupta N. Lower gastrointestinal bleeding in patients with coronary artery disease on antithrombotics and subsequent mortality risk. *J Gastroenterol Hepatol* 2018;33:1185-91.
- Laursen SB, Oakland K, Laine L, et al. ABC score: a new risk score that accurately predicts mortality in acute upper and lower GI bleeding: an international multicentre study. *Gut* 2021;70:707-16.
- Almaghrabi M, Gandhi M, Guizzetti L, et al. Comparison of risk scores for lower gastrointestinal bleeding: a systematic review and meta-analysis. *JAMA Netw Open* 2022;5:e2214253.
- Hreinsson JP, Sigurdardottir R, Lund SH, et al. The SHA(2)PE score: a new score for lower gastrointestinal bleeding that predicts low-risk of hospital-based intervention. *Scand J Gastroenterol* 2018;53:1484-9.
- Kherad O, Restellini S, Martel M, et al. Outcomes following restrictive or liberal red blood cell transfusion in patients with lower gastrointestinal bleeding. *Aliment Pharmacol Ther* 2019;49:919-25.
- Laine L, Shah A. Randomized trial of urgent vs. elective colonoscopy in patients hospitalized with lower GI bleeding. *Am J Gastroenterol* 2010;105:2636-41; quiz 2642.
- Tomaselli GF, Mahaffey KW, Cuker A, et al. 2020 ACC expert consensus decision pathway on management of bleeding in patients on oral anticoagulants: a report of the American College of Cardiology Solution Set Oversight Committee. *J Am Coll Cardiol* 2020;76:594-622.
- Coccolini F, Coimbra R, Ordonez C, et al. Liver trauma: WSES 2020 guidelines. *World J Emerg Surg* 2020;15:24.
- Aoki T, Nagata N, Yamada A, et al. Next endoscopic approach for acute lower gastrointestinal bleeding without an identified source on colonoscopy: upper or capsule endoscopy? *Endosc Int Open* 2019;7:E337-e346.
- Thiebaud PC, Yordanov Y, Galimard JE, et al. Suspected lower gastrointestinal bleeding in emergency departments, from bleeding symptoms to diagnosis. *Am J Emerg Med* 2019;37:772-4.
- Jensen DM, Machicado GA. Diagnosis and treatment of severe hematochezia. The role of urgent colonoscopy after purge. *Gastroenterology* 1988;95:1569-74.
- Lecleire S, Iwanicki-Caron I, Di-Fiore A, et al. Yield and impact of emergency capsule enteroscopy in severe obscure-overt gastrointestinal bleeding. *Endoscopy* 2012;44:337-42.
- Jacovides CL, Nadolski G, Allen SR, et al. Arteriography for lower gastrointestinal hemorrhage: role of preceding abdominal computed Tomographic angiogram in diagnosis and localization. *JAMA Surg* 2015;150:650-6.
- Green BT, Rockey DC, Portwood G, et al. Urgent colonoscopy for evaluation and management of acute lower gastrointestinal hemorrhage: a randomized controlled trial. *Am J Gastroenterol* 2005;100:2395-402.
- Lim DS, Kim HG, Jeon SR, et al. Comparison of clinical effectiveness of the emergent colonoscopy in patients with hematochezia according to the type of bowel preparation. *J Gastroenterol Hepatol* 2013;28:1733-7.
- Trastulli S, Barillaro I, Desiderio J, et al. Colonic explosion during treatment of radiotherapy complications in prostatic cancer. *Oncol Lett* 2012;4:915-8.
- Takasu A, Ikeya T, Shiratori Y. Comparison of conventional and new endoscopic band ligation devices for colonic diverticular bleeding. *Clin Endosc* 2022;55:408-16.
- Kobayashi K, Nagata N, Furumoto Y, et al. Effectiveness and adverse events of endoscopic clipping versus band ligation for colonic diverticular hemorrhage: a large-scale multicenter cohort study. *Endoscopy* 2022;54:735-44.

36. Honda H, Ishii N, Takasu A, et al. Risk factors of early rebleeding in the endoscopic management of colonic diverticular bleeding. *J Gastroenterol Hepatol* 2019;34:1784-92.
37. Kobayashi K, Furumoto Y, Akutsu D, et al. Endoscopic detachable snare ligation improves the treatment for colonic diverticular hemorrhage. *Digestion* 2020;101:208-16.
38. Kawano K, Takenaka M, Kawano R, et al. Efficacy of over-the-scope clip method as a novel hemostatic therapy for colonic diverticular bleeding. *J Clin Med* 2021;10:2891.
39. Kate V, Sureshkumar S, Gurushankari B, et al. Acute upper non-variceal and lower gastrointestinal bleeding. *J Gastrointest Surg* 2022;26:932-49.
40. Ng JL, Marican M, Mathew R. Topical haemostatic powder as a novel endoscopic therapy for severe colonic diverticular bleeding. *ANZ J Surg* 2019;89:E56-e60.
41. Jensen DM, Ohning GV, Kovacs TO, et al. Doppler endoscopic probe as a guide to risk stratification and definitive hemostasis of peptic ulcer bleeding. *Gastrointest Endosc* 2016;83:129-36.
42. Jensen DM, Ohning GV, Kovacs TO, et al. Natural history of definitive diverticular hemorrhage based on stigmata of recent hemorrhage and colonoscopic Doppler blood flow monitoring for risk stratification and definitive hemostasis. *Gastrointest Endosc* 2016;83:416-23.
43. Jensen DM. The importance of arterial blood flow detection for risk stratification and eradication to achieve definitive hemostasis of severe non-variceal UGI hemorrhage. *J Clin Med* 2023;12:6473.
44. Rodríguez de Santiago E, Hernández-Tejero M, Rivero-Sánchez L, et al. Management and outcomes of bleeding within 30 days of colonic polypectomy in a large, real-life, multicenter cohort study. *Clin Gastroenterol Hepatol* 2021;19:732-42.e6.
45. Sonnenberg A. Management of delayed postpolypectomy bleeding: a decision analysis. *Am J Gastroenterol* 2012;107:339-42.
46. Parra-Blanco A, Kaminaga N, Kojima T, et al. Hemoclippping for postpolypectomy and postbiopsy colonic bleeding. *Gastrointest Endosc* 2000;51:37-41.
47. Hookey L, Barkun A, Sultanian R, et al. Successful hemostasis of active lower GI bleeding using a hemostatic powder as monotherapy, combination therapy, or rescue therapy. *Gastrointest Endosc* 2019;89:865-71.
48. Nishimura N, Mizuno M, Shimodate Y, et al. Risk factors for active bleeding from colonic angiodysplasia confirmed by colonoscopic observation. *Int J Colorectal Dis* 2016;31:1869-73.
49. Olmos JA, Marcolongo M, Pogorelsky V, et al. Long-term outcome of argon plasma ablation therapy for bleeding in 100 consecutive patients with colonic angiodysplasia. *Dis Colon Rectum* 2006;49:1507-16.
50. Sriram N, Bar-Yishay I, Kumarasinghe P, et al. Definitive therapy of colonic angioectasia by submucosal coagulation. *Endosc Int Open* 2019;7:E1773-7.
51. Sato Y, Takayama T, Sagawa T, et al. Argon plasma coagulation treatment of hemorrhagic radiation proctopathy: the optimal settings for application and long-term outcome. *Gastrointest Endosc* 2011;73:543-9.
52. ASGE Standards of Practice Committee; Lee JK, Agrawal D, Thosani N, et al. ASGE guideline on the role of endoscopy for bleeding from chronic radiation proctopathy. *Gastrointest Endosc* 2019;90:171-82.e1.
53. Dohmoto M, Hünnerbein M, Schlag PM. Palliative endoscopic therapy of rectal carcinoma. *Eur J Cancer* 1996;32a:25-9.
54. Chen YI, Wyse J, Lu Y, et al. TC-325 hemostatic powder versus current standard of care in managing malignant GI bleeding: a pilot randomized clinical trial. *Gastrointest Endosc* 2020;91:321-8.e1.
55. Gayer C, Chino A, Lucas C, et al. Acute lower gastrointestinal bleeding in 1,112 patients admitted to an urban emergency medical center. *Surgery* 2009;146:600-6; discussion 606-7.
56. Köhler G, Koch OO, Antoniou SA, et al. Relevance of surgery after embolization of gastrointestinal and abdominal hemorrhage. *World J Surg* 2014;38:2258-66.
57. Loftus TJ, Brakenridge SC, Croft CA, et al. Neural network prediction of severe lower intestinal bleeding and the need for surgical intervention. *J Surg Res* 2017;212:42-7.
58. Pannatier M, Duran R, Denys A, et al. Characteristics of patients treated for active lower gastrointestinal bleeding detected by CT angiography: interventional radiology versus surgery. *Eur J Radiol* 2019;120:108691.
59. Sue-Chue-Lam C, Castelo M, Baxter NN. Factors associated with mortality after emergency colectomy for acute lower gastrointestinal bleeding. *JAMA Surg* 2020;155:165-7.
60. Chan S, Pittayanon R, Wang HP, et al. Use of over-the-scope clip (OTSC) versus standard therapy for the prevention of rebleeding in large peptic ulcers (size ≥ 1.5 cm): an open-labelled, multicentre international randomised controlled trial. *Gut* 2023;72:638-43.
61. Soriani P, Biancheri P, Bonura GF, et al. Over-the-scope clip as first-line treatment of peptic ulcer bleeding: a multicenter randomized controlled trial (TOP Study). *Endoscopy* 2024;56:665-73.

Abbreviations: AEs, adverse events; APC, argon plasma coagulation; ASGE, American Society for Gastrointestinal Endoscopy; GRADE, Grading of Recommendations Assessment, Development, and Evaluation; LGIB, lower GI bleeding; OR, odds ratio; RCT, randomized controlled trial; SMD, standardized mean difference; SOP, Standards of Practice; UGIB, upper GI bleeding.

© 2026 American Society for Gastrointestinal Endoscopy. Published by Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

0016-5107/\$36.00

<https://doi.org/10.1016/j.gie.2019.04.231>

Received June 11, 2026. Accepted June 12, 2026.

Current affiliations: Division of Gastroenterology and Hepatology, Department of Medicine, University of Calgary, Calgary, Alberta, Canada (1), Department of Community Health Sciences, University of Calgary, Calgary, Alberta, Canada (2), Department of Gastroenterology, Mayo Clinic, Rochester, Minnesota, USA (3), Cumming School of Medicine, University of Calgary, Calgary, Alberta, Canada (4), Department of Medicine, University of Calgary, Calgary, Alberta, Canada (5), Department of Surgery, University of Calgary, Calgary, Alberta, Canada (6), Division of Interventional Radiology, Department of Radiology, University of California at Los Angeles, Los Angeles, California, USA (7), Department of Radiology, University of Calgary, Calgary, Alberta, Canada (8), Section of Gastroenterology and Hepatology, Baylor College of Medicine, Houston, Texas, USA (9), Center for Interventional Endoscopy, AdventHealth, Orlando, Florida, USA (10), Center for Interventional Gastroenterology at UTHealth, McGovern Medical School, Houston, Texas, USA (11), Division of Gastroenterology and Hepatology, Massachusetts General Hospital, Boston, Massachusetts, USA (12), Department of Gastroenterology, Northwell, New Hyde Park, New York, USA (13), Division of Gastroenterology and Hepatology, University of Michigan, Ann Arbor, Michigan, USA (14), Division of Gastroenterology, Walter Reed Medical Center, Bethesda, Maryland, USA (15), Digestive Diseases Division, Uniformed Services University, Bethesda, Maryland, USA (16), Division of Gastroenterology and Hepatology, UMass Chan Medical School, Worcester, Massachusetts, USA (17), Division of Gastroenterology and Hepatology, Johns Hopkins University, Baltimore, Maryland, USA (18), Division of Pediatric Gastroenterology, Hepatology and Nutrition, Baylor College of Medicine, Texas Children's Hospital, Houston, Texas, USA (19), Center for Digestive Health, Penn Medicine Princeton Medical Center, Plainsboro, New Jersey, USA (20), Division of Gastroenterology, Beth Israel Deaconess Medical Center, Harvard Medical School, Boston, Massachusetts, USA (21), Division of Gastroenterology and Hepatology, Loma Linda University, Loma Linda, California, USA (22), Department of Gastroenterology, Hepatology, and Nutrition, University of Florida, Gainesville, Florida, USA (23).

Corresponding author: Nirav C. Thosani, MD, MHA, Center for Interventional Gastroenterology at UTHealth McGovern Medical School, 6431 Fannin Street, Houston, Texas 77030, USA. E-mail: nirav.thosani@uth.tmc.edu.